

NITROGEN DIFFUSION IN BCC IRON

M. L. Wasz and Rex B. McLellan
Department of Mechanical Engineering and Materials Science
William Marsh Rice University, Houston, Texas 77251-1892

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Introduction

The availability of different measuring techniques has enabled the diffusivity D of C and N in bcc Fe to be determined over ~ 15 orders of magnitude. It has generally been concluded that in both cases the diffusivity data could not be adequately represented by a single Arrhenius equation. Numerous models have been set forth in order to explain the apparent curvature observed in the Arrhenius plot of the values of $\ln D$ vs. $1/T$. The most cogent model, originally proposed by Wuttig (1), ascribes the curvature to the ferromagnetic transformation in Fe. The effect of this transformation on self-diffusion in Fe is well known (2) and is clearly observed in the diffusivity of substitutional elements in Fe which diffuse by a vacancy exchange mechanism (3).

Recent internal friction measurements on C and N diffusion by Weller (4), in an intermediate temperature range, have however been interpreted to mean that in fact all the diffusion data may be described by a single linear Arrhenius plot. The C-diffusivity data have been re-examined (5) and it seems clear that the high temperature data generated by mass flow techniques, above the Curie temperature, are in excellent mutual accord and are about five times greater than the extrapolated results of Weller. The purpose of the present communication is to re-examine the data for N-diffusion in bcc iron in the light of Weller's contentions.

N-Diffusivity Data

A compilation of the N-diffusivity is presented in Fig. (1). The different authors and experimental techniques are given in the symbol-author code below.

Author	Symbol	Method	Author	Symbol	Method
6	△	IF	13	◇	MF
7	▽	IF	14	▼	IF
8	▲	IF	15	+	IF
9	●	MPD	16	◆	IF
10	○	IF	17	■	IF, MF
11	■	IF	18	□	IF
12	□	MF			

The abbreviations used are: IF = internal friction, MF = mass flow, and MPD = magnetic permeability disaccommodation. Note that Fast and Verrijp (17) used IF at low temperatures (282.5 and 294.5 K) and MF at higher temperatures.

The data, using the symbol code given above, are shown in Arrhenius representation in Fig. (1). The solid line is a least squares regression to the data below $10^4/T = 29$. The relation is given by,

$$\ln D = -0.861 (10^4/T) - 16.763 \quad \text{Eqn. (1)}$$

where D is in m^2/sec and the correlation coefficient is $R^2 = 0.998$.

Discussion

The diffusivity data are plotted in the low and high temperature regimes in separate plots in Fig. (2). The uppermost plot (low temperature) utilizes the lower reciprocal temperature scale. The dashed lines in Fig. (2) represent extrapolations of the result given by Weller (4),

$$\ln D = -0.9505 (10^4/T) - 13.867 \quad \text{Eqn. (2)}$$

The data upon which Eqn. (2) is based were measured using an iron doped with 40 at. ppm of N and the temperature range of ~ 285 to 400 K. Fig. (2) shows that Weller's data deviate from the low temperature data as the data region corresponding to the lowest temperatures are approached. It is also apparent that the slope of Eqn. (2) does not correspond to the plot of the high temperature data depicted in Fig. (2).

It is unfortunate that the diffusivity data in the δ -Fe region (13) are relatively scant. This fact, coupled with the short span of $1/\Delta T$ compared with ΔT as T increases, makes it difficult to evaluate the diffusion data in the light of magnetic ordering, as has been done for substitutional solutes (3). It is, however, worthwhile pointing out that curvature observed in the diffusion of C in Co does not appear to be related to the magnetic disorder transformation (19), but by the same token the large anomaly found for self-diffusion in Fe at the Curie temperature (2) is not seen in Co self-diffusion (20, 21).

Acknowledgments

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Fig. (1) – Arrhenius plot of $\ln D$ vs $1/T$ for the diffusion of N in bcc Fe. The author-symbol code is given in the text.

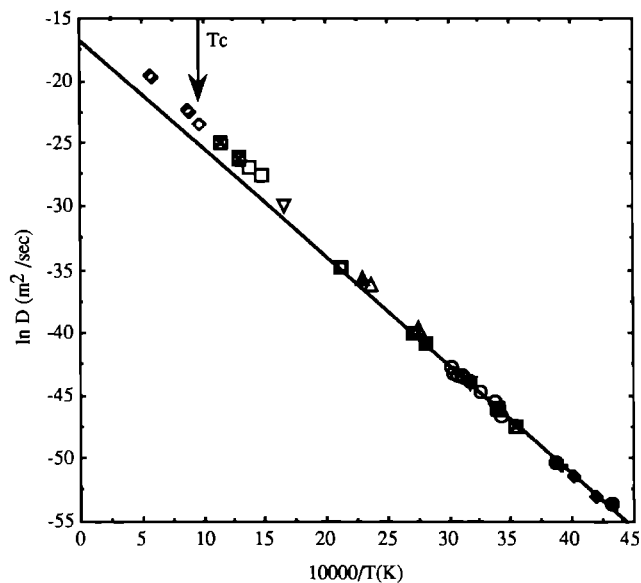


Fig. (2) – Arrhenius plots for the diffusion of N in bcc Fe separated into high and low temperature regimes. The dashed line is Weller's extrapolation.

